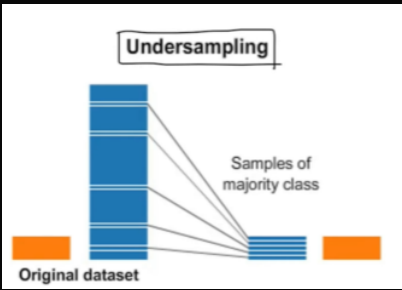
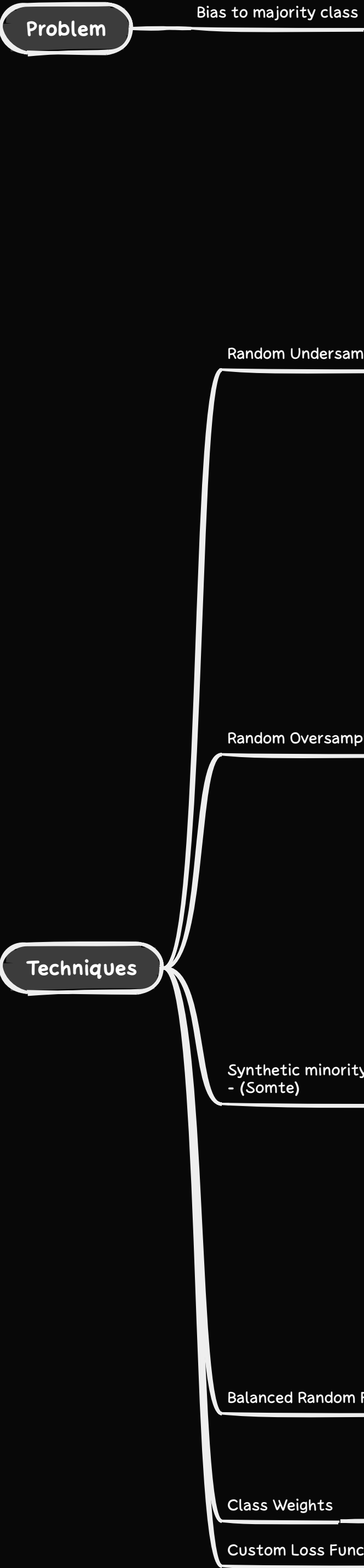
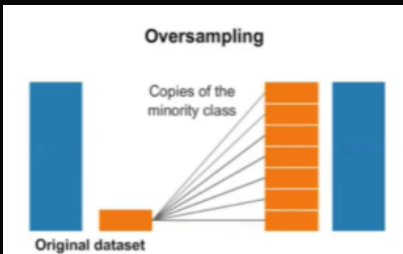


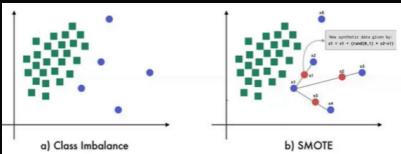
ImBalanced-Data



Reducing the size of the majority class to make it equal to minority class



Increasing the size of minority class by adding duplicate rows of minority class



Here we also -> increase the size of minority class but not by adding duplicate rows. Instead of generating new points.

Not able to work good on categorical dataset [x is categorical columns]

```
from imblearn.over_sampling import SMOTE
smote = SMOTE(random_state=42)
X_resampled_smote, y_resampled_smote = smote.fit_resample(X_train, y_train)
```

```
from imblearn.ensemble import BalancedRandomForestClassifier
classifier = BalancedRandomForestClassifier(random_state=42)
classifier.fit(X_train, y_train)
```

Random forst but for imbalanced dataset

Algo support weights as hyper parameter to increase the weight of minority class

Custom Loss Function